

POKHARA UNIVERSITY

Level: Bachelor
Programme: BCSIT
Course: Digital System

Semester: Fall

Year: 2025
Full Marks: 100
Pass Marks: 45
Time: 3 hrs.

Candidates are required to answer in their own words as far as practicable. The figures in the margin indicate full marks.

Section "A"

Very Short Answer Questions

1. Attempt all the questions. [10×2=20]
1. Explain ASCII with example.
2. Mention the drawback of analog system over digital system.
3. What do you mean by universal gates? Why?
4. Explain duality principle with one example.
5. What is a modulo-n counter?
6. What is multiplexer? How many select lines are required in 16 X 1 multiplexer.
7. What is ROM?
8. How many flip flops are required to generate a 1 Hz pulse from 64 Hz pulse.
9. Define one bit memory.
10. What do you mean by triggering of flip flop?

Section "B"

Descriptive Answer Questions

11. Attempt any five questions. [5×10=50] 30
11. Perform the following conversions:
 - a. $(10111001)_2 = (?)_{10}$ excess-3 code
 - b. $(1256.7)_8 = (?)_9$
 - c. $(18129.24)_{10} = (?)_8$
 - d. $(AD5B.C34)_{16} = (?)_{10}$
 - e. $(1567.24)_8 = (?)_{16}$
12. Design and explain a full subtractor logic circuit using two half subtractor.
 - a. Design a logic circuit with 3 inputs, A, B, and C will have its output HIGH only when most of the inputs are low.
13. Simplify the following expression.

$$F = \sum_m (0,2,4,6,11,15) + \sum_d (8,10,12,13,14)$$
 Using k-map and draw the logic diagram using
 - a. Basic gates only
 - b. NAND gates only

14. Implement the following function $\Sigma (0,1,3,4,8,9,10)$ using
 - a. Decoder
 - b. Multiplexer

15. What is a synchronous counter. Design a synchronous counter with counting sequences: 000, 001, 011, 111, 110, 100, 000.
16. What is master slave flip-flop? Design SR flip-flop logic circuit and explain the working principle.

Group "C"

Long Answer Questions

17. Attempt any two questions. [2×15=30]
17. Differentiate between PLA and PAL. Realize the following Boolean expression using PLA and PAL.
 - a. $BD + A'BC' + ABC' + ABC$
18. Create a sequence detector that detects the sequence 1010 using T flip flop.
19. Design a sequential circuit from following state diagram using basic gates and JK flipflop.

